

Channel Independence Wins All?

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Motivation:

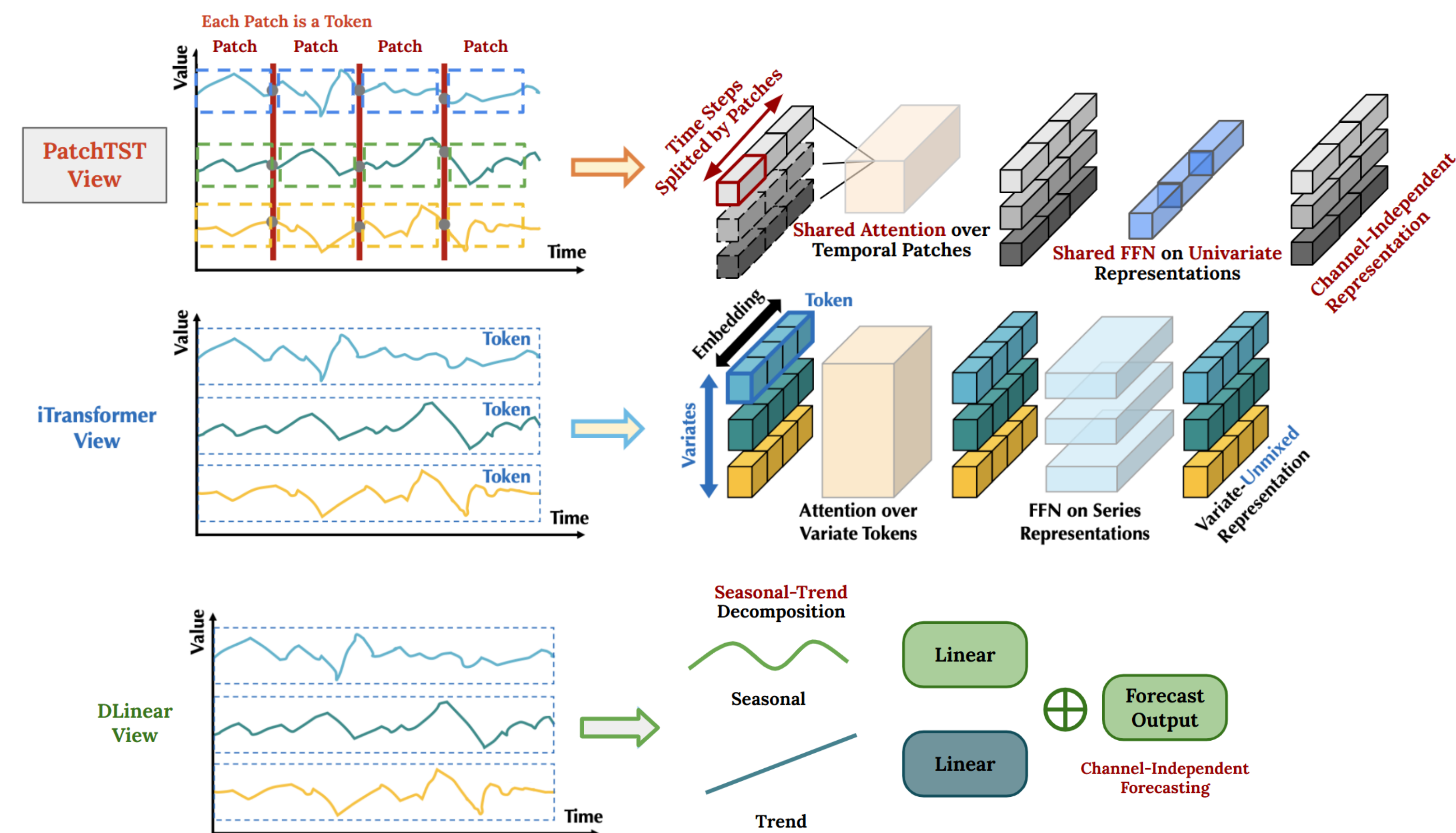
Two recent transformer-based LTSF (long-term time series forecasting) proposals offer opposing inductive biases:

- PatchTST (ICLR 2023): time-axis tokens with Channel Independence (CI). Each variable forecasted in isolation.
- iTransformer (ICLR 2024): variable-axis tokens with Channel Dependence (CD). Cross-channel attention is the core.

Both papers claim superiority on multivariate benchmarks.

We ask: under matched compute budget , with each method at its recommended configuration — which architecture wins?

Architecture Comparison:



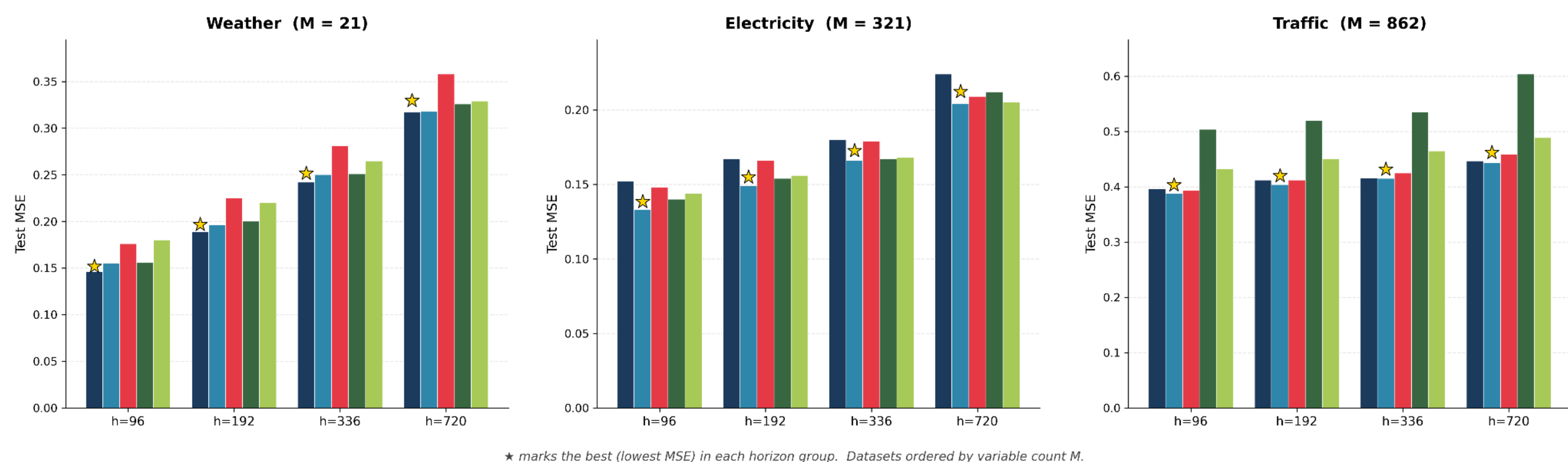
Experimental Setup & Reproduction quality:

								Reproduction quality — Δ% vs published results												
Model	Lookback	Patch	Stride	N	Layers	d_model	Source	Model	Weather				Electricity				Traffic			
PatchTST/64	512	16	8	64	3	128	Custom impl.	PatchTST/64	9%	19%	33%	72%	9%	19%	33%	72%	9%	19%	33%	72%
PatchTST/42	336	16	8	42	3	128	Custom impl.	PatchTST/42	-2.0%	-2.6%	-1.2%	+1.0%	+17.8%	+13.6%	+10.4%	+13.7%	+10.0%	+8.7%	+6.1%	+3.5%
iTransformer	96	—	—	—	3-4	512	Official iTrans repo	iTransformer	+2.0%	-0.5%	+0.4%	-0.6%	+2.3%	+0.7%	-0.6%	+1.0%	+5.7%	+4.9%	+4.3%	+2.3%
DLinear (l / shared)	336	—	—	—	—	—	Official PatchTST repo	DLinear	+1.1%	+1.8%	+1.1%	0.0%	0.0%	+2.5%	+0.6%	-7.1%	-0.5%	-1.2%	-1.8%	-1.7%
All models trained for 20 epochs. Each uses its paper-recommended lookback. $N = \lfloor (L - P) / S \rfloor + 2$								Green = within paper range · Yellow = moderate deviation · Red = significant deviation (undertrained)												

Main Results:

PatchTST achieves the lowest MSE on 12/12 (dataset, horizon) cells — under matched 20-epoch compute

Legend: PatchTST/64 (dark blue), PatchTST/42 (medium blue), iTransformer (red), DLinear-l (dark green), DLinear (light green)



★ marks the best (lowest MSE) in each horizon group. Datasets ordered by variable count M.

Reference:

- Nie, Yuqi, et al. "A time series is worth 64 words: Long-term forecasting with transformers." arXiv preprint arXiv:2211.14730 (2022).
- Liu, Yong, et al. "itransformer: Inverted transformers are effective for time series forecasting." arXiv preprint arXiv:2510.06625 (2025).
- Zeng, Ailing, et al. "Are transformers effective for time series forecasting?." Proceedings of the AAAI conference on artificial intelligence. Vol. 37. No. 9. 2023.

Comparison with No-Attention based Model

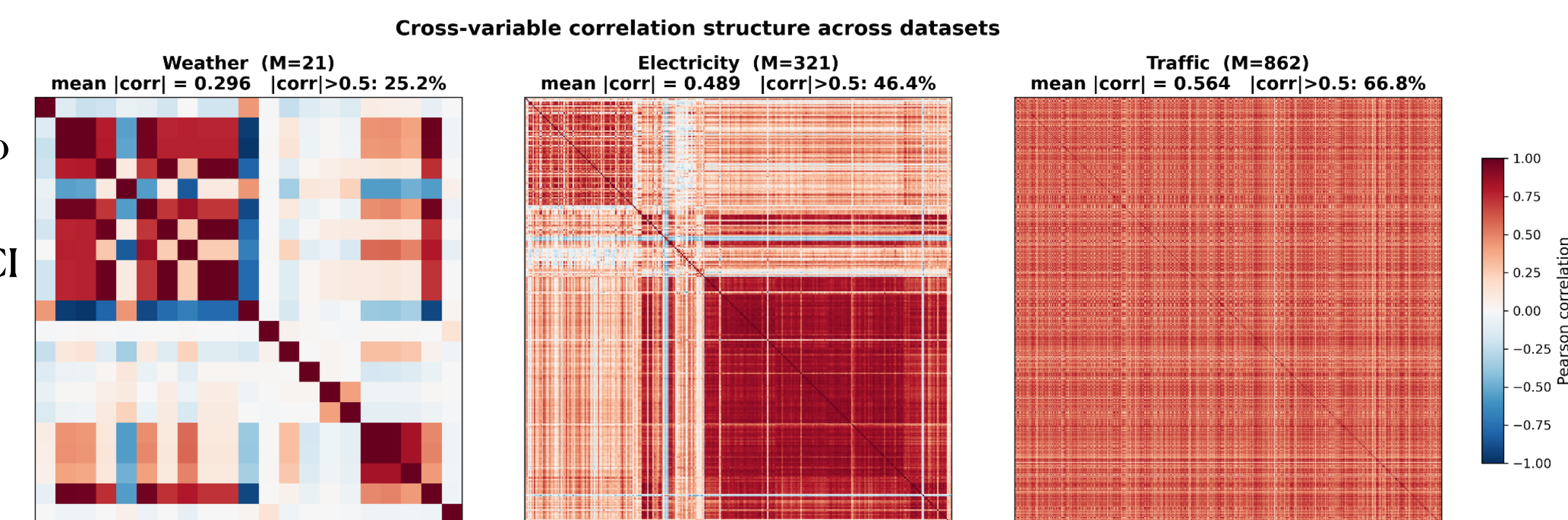
- DLinear decomposes a time series into trend and seasonal components using a moving average kernel, then passes each through a separate single linear layer and sums the outputs - matching or beating Transformer-based models at a fraction of the complexity.

Dataset	M	h	With Attention			Without Attention		
			PatchTST/42 (CI, attn)	iTransformer (CD, attn)	Δ attn (%)	DLinear-l (CI, no attn)	DLinear (CD, no attn)	Δ no attn (%)
Weather	21	96	0.155	0.176	+11.9	0.156	0.180	+15.4
Weather	21	192	0.196	0.225	+12.9	0.200	0.220	+10.2
Weather	21	336	0.250	0.281	+11.0	0.251	0.265	+5.4
Weather	21	720	0.318	0.358	+11.2	0.326	0.329	+1.0
Electricity	321	96	0.133	0.148	+10.1	0.140	0.144	+2.9
Electricity	321	192	0.149	0.166	+10.2	0.154	0.156	+1.2
Electricity	321	336	0.166	0.179	+7.3	0.167	0.168	+0.8
Electricity	321	720	0.204	0.209	+2.4	0.212	0.205	-3.2
Traffic	862	96	0.388	0.393	+1.3	0.504	0.432	-14.3
Traffic	862	192	0.404	0.412	+1.9	0.520	0.451	-13.1
Traffic	862	336	0.415	0.425	+2.4	0.535	0.465	-13.0
Traffic	862	720	0.444	0.459	+3.3	0.604	0.489	-19.1

Δ = relative MSE reduction of CI over CD (positive = CI wins). Bold = lower MSE in each pair.

When does CI Work?

- For our datasets the correlation increases as the size of the dataset increases. We also see that the gap between CM models (iTransformer and original DLinear) and CI models decreases as correlation increases.
- CI works better when the channels have distinguished characteristics (low correlation). CM looks better when the channels are more correlated.

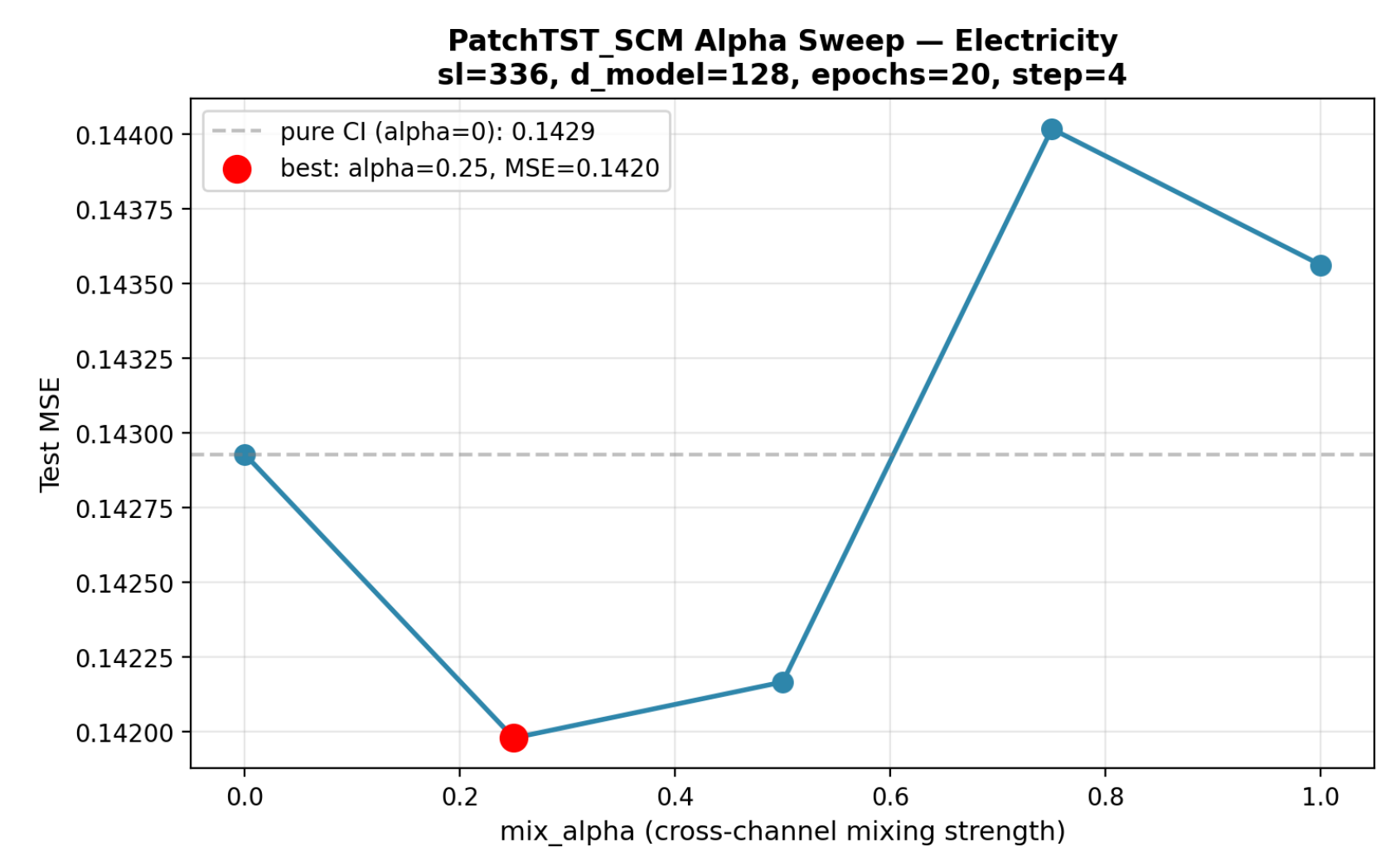


Lets add some Channel Mixing...But how? Stochastic CM

Intuitively, as M grows and cross-variate correlation increases, channel mixing should help more. The question is not whether to mix, but how much.

We add a lightweight linear mixer on top of PatchTST's CI output with interpolation weight α ($\alpha=0$: pure CI; $\alpha=1$: full mixing).

Result: Performance follows a U-shaped curve — optimal at $\alpha=0.25$ (-0.6% MSE), but $\alpha \geq 0.75$ degrades below pure CI. Less mixing, better generalization.



Conclusion & Contribution:

Under matched 20-epoch compute across 3 benchmarks (Weather, Electricity, Traffic) and 4 forecast horizons:

- PatchTST (CI) achieves the lowest MSE on 12/12 cells. iTransformer (CM) never wins.
- **CI is a safe default, with room for controlled mixing on highly correlated data:** CI wins when channels are distinct; as correlation grows, CI's edge shrinks but never disappears.
- **But we can do better with proper mixing.** SCM shows heavy mixing hurts ($\alpha \geq 0.75$), yet light mixing ($\alpha=0.25$) yields marginal gains — the future is CI-first with adaptive, dataset-aware mixing.

Future Work

- **Computational robustness.** Extend to full training budgets (100 epochs) and multi-seed evaluation to test whether CI's advantage holds
- **SCM extension.** Sweep the mixing parameter α across all datasets and horizons to map dataset-specific optimal CI/CM blending
- **Broader benchmarks.** Evaluate on Solar, PEMS, and ETT to characterize when CI vs CM is most beneficial, moving toward dataset-aware architecture selection.

